AS1202 – Introduction to VBA for Excel

Group Coursework: Index-Linked Bond Cash Flow Model with Interactive User Form VBA/Excel Program

USER GUIDE

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Introduction ----------

The Index Linked Bond Cash Flow Model was created to be used as a tool to aid investors in their decision if a given bond is a worthwhile financial investment,

accounting for all possible taxes and inflationary effects. It allows calculation of a cash flow table at every point of the coupon distribution, as well as a yield estimation that is done using linear interpolation to calculate it to an accuracy of 4 decimal places.

For the bonds to be analysed, they must fit the following requirements:

- It must be purchased between January 2001 and December 2024
- It must have at least 10 years, but no more than 35 years, before the term of the bond reaches maturity.

Section 1 – User Form ----------

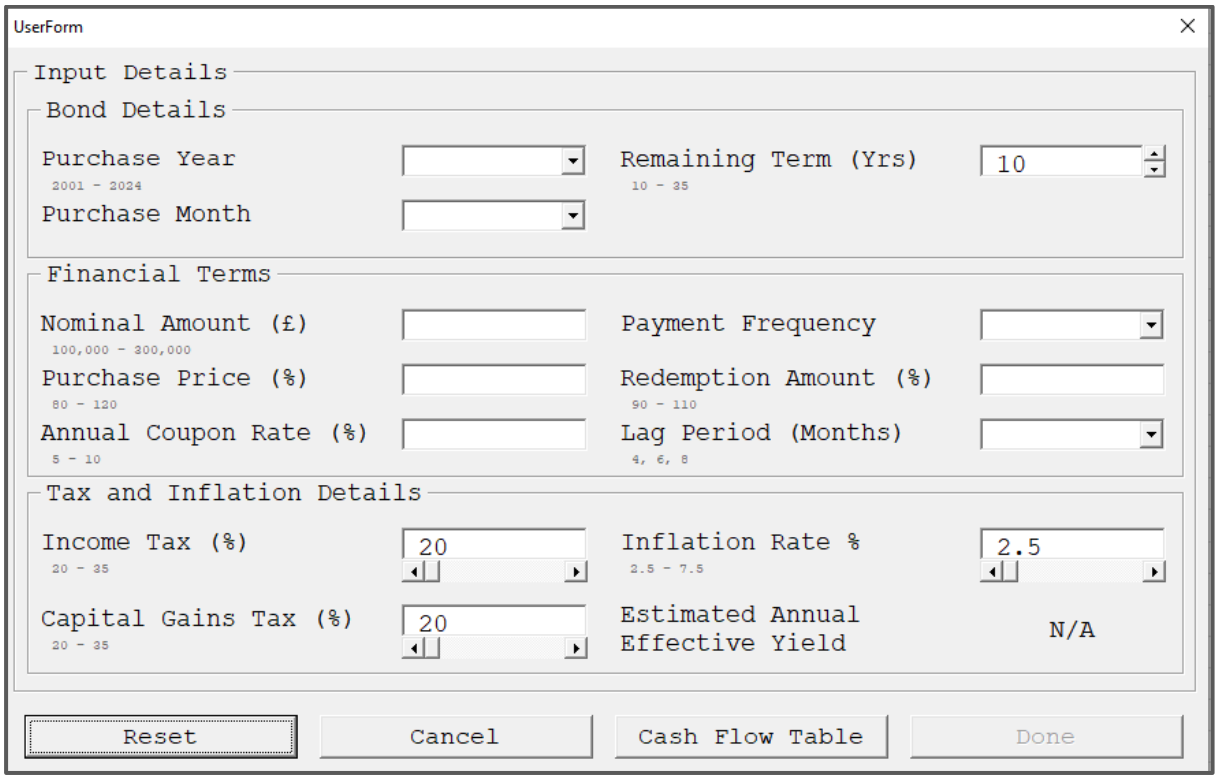

Figure 1: User Form in Default State

The User Form takes advantage of a three-step process of Setup, Calculation, and Completion, intended to guide users through the process of pricing a bond in a systematic order.

Inputs of the User Form ----------

Inputs	Explanation	Parameters
Purchase Year/Month	The month and year of which the bond was purchased	Year: 2001-2024 Month: January-December
Remaining Term (Years)	How long until the bond matures	Integer Between 10-35
Nominal Amount Purchased	Face value of the bond multiplied by number of bonds purchased	£100 000-300 000
Purchase Price	Amount of money spent on the purchase of the bond (as a percentage of nominal amount)	80-120%
Annual Coupon Rate	Coupons given as a percentage of the nominal purchased amount	5-10%
Payment Frequency	How often coupons are paid out to the bondholder	Monthly (12x/Year) Quarterly (4x/Year) Half-Yearly (2x/Year) Yearly (1x/Year)
Redemption Amount	Percentage of nominal amount that is paid at maturity of the bond	90-110%
Lag Period	The number of months before inflation affects returns	4 Months 6 Months 8 Months
Income and Capital Gains Tax	Taxes levied on the price of the bond (Capital Gains Tax only applicable if redemption price is higher than purchase price)	20-35%
Inflation Rate	Projected future inflation	2.5-7.5%

If an input is incomplete or otherwise wrong, the user form will not be able to confirm the data and the necessary outputs will not be available for visibility until such mistakes are rectified.

There are also buttons and other quality of life improvements in the form of secondary controls to increase accessibility and convenience of use.

Buttons ----------

Button	Function

Reset	Set all input boxes to blank (20% for Income & Capital Gains Tax, 2.5% for Inflation, and 10 for length until maturity) as seen in the above image
Cancel	Close the User Form without saving changes
Cash Flow Table	Generate the Cash Flow Table and outputs the yield, as well as ensure all inputs are valid.
Done	Confirm all values and saves them, as well as validates all inputs

Secondary Controls ----------

- Scroll Bar →
  - Allows for changes in values in increments of 0.1% by clicking on the left and right buttons, and 0.5% by dragging the scroll bar.
  - Applicable to:
    - Income Tax
    - Capital Gains Tax
    - Inflation

- Drop-Down Menus →
  - Allows for users to pick between a set of predetermined values rather than inputting their own, minimising the risk of errors.
  - Applicable to:
    - Purchase Year
    - Purchase Month
    - Remaining Term
    - Payment Frequency
    - Lag Period

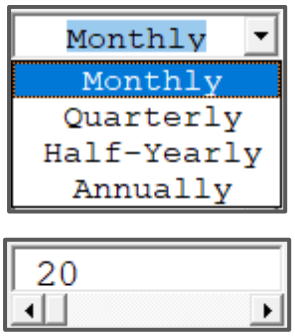

Figure 2 and 3: Example of a Drop-Down Menu and Scroll Bar

Section 2 – Outputs ----------

Cash Flow Tables Components ----------

Component	Explanation
Payment Date	Date at which coupons are paid out
Lagged Index	Inflation Rate as determined by CPI data/given inflation in the user form (see below)
Index Ratio	Lagged Index of the current year represented as a ratio to the base year (first year of the bond)
Base Coupon Amount	Amount paid out per coupon payment
Indexed Coupon Amount	Amount paid out per coupon payment accounting for the index Ratio
Taxed Coupon Amount	Indexed coupon amount accounting for income tax
Capital Gains Tax	Tax that gets levied on the redemption price of the bond
Net Cash Flow	The amount of cash that gets added to or subtracted from the overall balance at the previous time of coupon payment
Balance	Overall balance at the end of the time of coupon payment

Cash Flow Table Example ----------

Payment Date	Lagged Index	Index Ratio	Base Coupon Amount	Indexed Coupon Amount	Taxed Coupon Amount	Redemption Amount	Capital Gains Tax	Net Cash Flow	Balance
Nov 2016	1021.3	1						-£264,600.00	-£264,600.00
May 2017	1025.8	1.004406149	£10,800.00	£10,847.59	£7,593.31	£0.00	£0.00	£7,593.31	-£257,006.69
Nov 2017	1045.8	1.023989034	£10,800.00	£11,059.08	£7,741.36	£0.00	£0.00	£7,741.36	-£249,265.33
May 2018	1060.6	1.038480368	£10,800.00	£11,215.59	£7,850.91	£0.00	£0.00	£7,850.91	-£241,414.42
Nov 2018	1083.6	1.061000685	£10,800.00	£11,458.81	£8,021.17	£0.00	£0.00	£8,021.17	-£233,393.26
May 2019	1091.7	1.068931754	£10,800.00	£11,544.46	£8,081.12	£0.00	£0.00	£8,081.12	-£225,312.13
Nov 2019	1108	1.084891805	£10,800.00	£11,716.83	£8,201.78	£0.00	£0.00	£8,201.78	-£217,110.35
May 2020	1122.9	1.099481054	£10,800.00	£11,874.40	£8,312.08	£0.00	£0.00	£8,312.08	-£208,798.27
Nov 2020	1145.8	1.121903456	£10,800.00	£12,116.56	£8,481.59	£0.00	£0.00	£8,481.59	-£200,316.68
May 2021	1160.7	1.136492705	£10,800.00	£12,274.12	£8,591.88	£0.00	£0.00	£8,591.88	-£191,724.80
Nov 2021	1186.6	1.161852541	£10,800.00	£12,548.01	£8,783.61	£0.00	£0.00	£8,783.61	-£182,941.19
May 2022	1189.6	1.164789974	£10,800.00	£12,579.73	£8,805.81	£0.00	£0.00	£8,805.81	-£174,135.38
Nov 2022	1202.2	1.177127191	£10,800.00	£12,712.97	£8,899.08	£0.00	£0.00	£8,899.08	-£165,236.30
May 2023	1223.948765	1.198422369	£10,800.00	£12,942.96	£9,060.07	£0.00	£0.00	£9,060.07	-£156,176.23
Nov 2023	1257.156856	1.23093788	£10,800.00	£13,294.13	£9,305.89	£0.00	£0.00	£9,305.89	-£146,870.34
May 2024	1291.265947	1.264335599	£10,800.00	£13,654.82	£9,558.38	£0.00	£0.00	£9,558.38	-£137,311.96
Nov 2024	1326.300483	1.298639463	£10,800.00	£14,025.31	£9,817.71	£0.00	£0.00	£9,817.71	-£127,494.24
May 2025	1362.285574	1.333874057	£10,800.00	£14,405.84	£10,084.09	£0.00	£0.00	£10,084.09	-£117,410.16
Nov 2025	1399.24701	1.370064633	£10,800.00	£14,796.70	£10,357.69	£0.00	£0.00	£10,357.69	-£107,052.47
May 2026	1437.211281	1.40723713	£10,800.00	£15,198.16	£10,638.71	£0.00	£0.00	£10,638.71	-£96,413.75
Nov 2026	1476.205596	1.445418188	£10,800.00	£15,610.52	£10,927.36	£0.00	£0.00	£10,927.36	-£85,486.39
May 2027	1516.257901	1.484635172	£10,800.00	£16,034.06	£11,223.84	£0.00	£0.00	£11,223.84	-£74,262.55
Nov 2027	1557.396903	1.524916189	£10,800.00	£16,469.09	£11,528.37	£0.00	£0.00	£11,528.37	-£62,734.18
May 2028	1599.652086	1.566290107	£10,800.00	£16,915.93	£11,841.15	£0.00	£0.00	£11,841.15	-£50,893.03
Nov 2028	1643.053733	1.608786579	£10,800.00	£17,374.90	£12,162.43	£0.00	£0.00	£12,162.43	-£38,730.61
May 2029	1687.632951	1.652436063	£10,800.00	£17,846.31	£12,492.42	£0.00	£0.00	£12,492.42	-£26,238.19
Nov 2029	1733.421688	1.697269841	£10,800.00	£18,330.51	£12,831.36	£0.00	£0.00	£12,831.36	-£13,406.83
May 2030	1780.452763	1.743320046	£10,800.00	£18,827.86	£13,179.50	£0.00	£0.00	£13,179.50	-£227.33
Nov 2030	1828.759881	1.790619682	£10,800.00	£19,338.69	£13,537.08	£483,467.31	£61,282.85	£435,721.55	**£435,494.22**

Figure 4: Example of a Cash Flow Table

Important Notes of Output Cash Flow Table ----------
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- Lagged Index as well as Index Ratio is taken from CPI Data for all periods of time before 2022. After 2022, projected Inflation Rate inputted in the User Form is used.
- If Balance is positive at the last listed time of coupon payment, or when the bond hits maturity and will be redeemed at the specified redemption amount, it is considered profits after all taxes.
- The Cash Flow Table example provided in the image was generated under the following parameters:

    - **Purchase Date**: November 2016
    - **Remaining Term**: 14 Years
    - **Nominal Amount**: 270000
    - **Payment Frequency**: Half-Yearly
    - **Purchase Price**: 98%
    - **Redemption Amount**: 100%
    - **Annual Coupon Rate**: 8%
    - **Lag Period**: 4 Months
    - **Income Tax**: 30%
    - **Capital Gains Tax**: 28%
    - **Inflation Rate**: 5.5%

Yield Calculator ----------

Estimated Annual Effective Yield	11.6217%

Figure 5: Example of the Estimated Annual Effective Yield Calculator

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The Estimated Annual Effective Yield calculates the effective yearly yield returns of the bond investment, and will be outputted on the User Form itself.

This is a great indicator for an investment to see if the yield is high enough to be worth the potential risk that could be undertaken from an investment in a bond.

This is most relevant for Junk Bonds and other forms of Corporate Bonds as they carry a higher level of volatility, and therefore a higher yield may be required to compensate for the level of volatility.

Section 3 – Instructions and Problems ----------

How to Use the Program

1. Open the Excel Software
    - Ensure Macros are enabled; the program will not work without it.
2. Enter Required Inputs as found in the User Form
    - Ensure that all inputs are within parameters and are all filled.
        i. To ensure that all inputs have values associated with them, the user will be unable to confirm inputs or generate the cash flow table without all inputs being filled.
        ii. To ensure all values are within specified parameters, the program will output error messages explaining the error and what the parameters required are.
3. Press 'Cash Flow Table' to generate the cash flow table from the given parameters, as well as output the yield of the investment.
    - If projected yield is above 20% or below 0%, the yield will not be generated, stating "TOO HIGH" or "TOO LOW" respectively.

Potential Problems ----------

- If an input is left empty, the 'Done' button will remain disabled until this is no longer the case.
- If an input is outside its respective range, a validation check will arise, outputting an error.
    - To rectify this, the input of the box with the error must be changed to that of a valid input within the designated parameters.

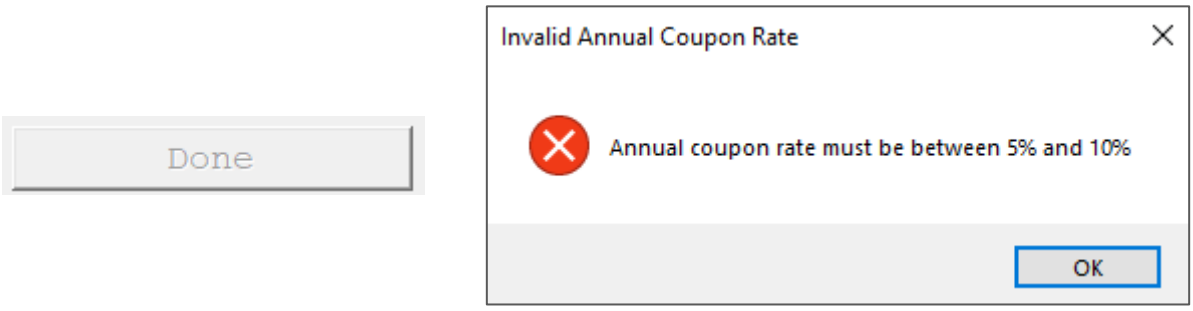

Figure 6 and 7: Example of Disabled Button and an Error Message From the User Form

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- The cash flow may not be entirely accurate to reality if parameters set differ from the real-world situation, investors utilizing this program shouldn't interpret the model's outputs as definitive or absolute.